

Micro Protection Fluid

ETF Style Adaptive Insurance for the Agentic Era
— GBA Market Penetration —

1. Executive Summary

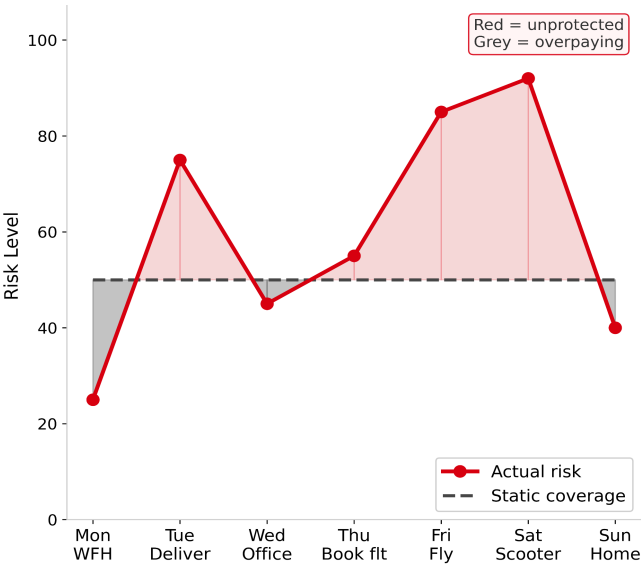
Our team proposes **Micro Protection Fluid**: an ETF-style adaptive insurance solution that addresses the static coverage problem in the agentic era. By combining agentic AI with real-time card authorization data, merchant category codes, and supplementary signals like weather and flight status, we aim to dynamically reallocate a fixed premium across risk categories for HSBC Life customers while creating new revenue opportunities in the GBA market.

2: Problem Statement & Opportunity

The problem: Insurance today is static. You pay a fixed premium, you get fixed coverage, and it never adjusts whether you are sitting at home or flying across Asia. You overpay for risks you do not face and underpay for risks you do.

Hong Kong and the Greater Bay Area:

Gen Z here works freelance Monday, delivers food Tuesday, flies to Taipei Friday. GBA cross-border mobility means your risk environment shifts between Hong Kong, Shenzhen, and Macau on a daily basis.



Why it matters now:

1. 9 trillion USD global protection gap
2. Below 1% insurance penetration across most of Asia
3. 78% of insurance executives see closing this gap as an ethical obligation

3: Proposed Solution

Core Idea: Insurance that works like an ETF with one fixed premium. AI reallocates it across coverage categories in real time based on what the customer is actually doing. Total cost never changes but the allocation changes.

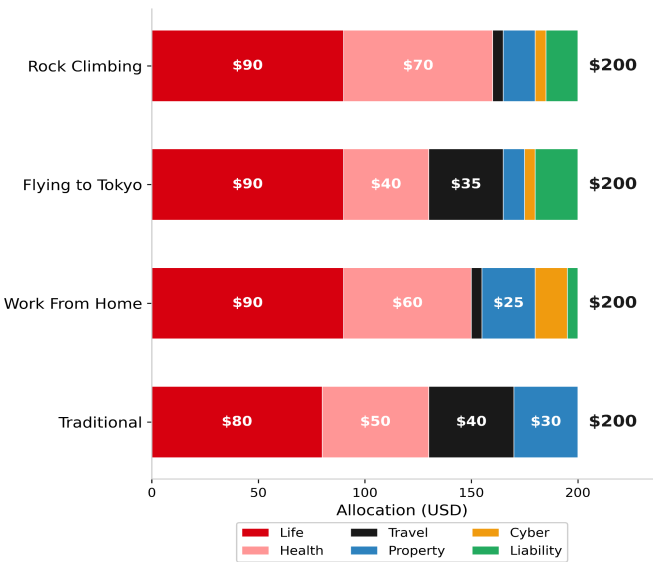
Key Features:

1. Fixed premium, variable depth: \$200/month flows to where risk is highest
2. Floor-and-ceiling model: Every category has a guaranteed minimum that AI cannot breach
3. Real-time triggers: AI reads card auth data (merchant codes, location, amount) within seconds
4. Full transparency: Customers see allocation live in the HSBC app
5. Customer control: Users set personal floors to lock in priorities

How It Works:

1. Toggle on in HSBC app, needing zero forms
2. AI sets baseline from spending history
3. Book a flight → AI detects airline merchant code → travel coverage increases instantly
4. Return home → allocation reverts to baseline
5. Activity exceeds coverage → optional micro-boost notification ("Boost for \$3.50. Tap to approve.")

Same \$200. Smarter Allocation.



Technology:

1. Sub-second response via card authorization data (not settlement)
2. Merchant Category Codes classify activity automatically
3. Supplementary signals: weather APIs, flight status, public event data
4. Agentic AI acting within guardrails, aligned with HSBC AI strategy

Integration with HSBC Life:

1. Upgrade for existing customers with zero acquisition cost
2. Uses HSBC's own transaction data of 41M customers and not third-party connections
3. Composite general insurance product alongside life policies
4. Fits HSBC 4Qs: Simple, Agile, Timeless, Personal

4. Strategy & Execution Plan

Go-to-Market Strategy: The Micro-Protection Fluid introduces insurance as a Function-as-a-Service (FaaS): Thousands of atomic "shields" that activate for minutes at a time, not years. Traditional insurance feels heavy and rigid, while this model delivers cover at \$20 per event, making it accessible, commitment-free, and triggered automatically by real-life risk.

Launch: It happens inside HSBC's existing mobile app (zero acquisition friction). Q4 2026 in Hong Kong and GBA cities, later expanding across Asia-Pacific via embedded channels.

Pricing: \$20 per micro-event. Agentic AI diverts tiny portions from long-term funds to power short-term shields, requiring no separate policy.

Profitability Table

Lever	Year 1 HK\$	Year 3 HK\$	Proof
Smarter Payouts	1.2B	3.6B	20pt loss ratio improvement (Straits Research' 25)
Zero Acquisition	480M	1.4B	8% saved
Churn Reduction	975M	2.9B	6% fewer leave x HKD 13k banking value (JDP' 25)
Reinsurance Save	144M	427M	15% on 16% ceded premium
TOTAL	2.8B	8.3B	

Target Customer Segment: Our target customer looks something like: Alex Rivera, 20-27-year-old urban professional (tech, consulting,

freelance), retail customer in the GBA area. Earns \$120k-\$200k per year, uses the HSBC app daily for ridesharing, short-haul travel, e-commerce, and gym.

Rejects rigid long-term insurance. Wants precise, on-demand protection without overpaying. Already shares spending/location data with HSBC, allowing agentic AI that acts faster than any manual process to buy the required insurance accordingly.

Partnerships:

Mobility & e-commerce: Didi, Uber, Trip.com, Alibaba → where it auto-activates shields for a stormy taxi ride or a single new-site transaction.

Health & activity: Hospitals + HSBC health-tech initiatives → short medical top-ups for gym sessions or weekend trips.

GBA entities: HSBC's Wealth Management Connect + mainland pilots → regulatory sandboxes for inclusive micro-insurance.

HSBC retains full ownership of customer relationships and data.

Phased Implementation Roadmap

Phase 1: Pilot (HK only)	Months 1-6	10,000 HSBC retail customers. Card-auth integration, flight/MCC triggers. Manual guardrails.
Phase 2: GBA Launch	Months 7-12	Expand to Macau and GBA account holders. Add weather + flight status APIs for claims. Launch micro-boost notifications.
Phase 3: Agentic Scale	Months 13-18	Full agentic AI (auto-rebalancing within floors). Regulatory sign-off for cross-border parametric payouts.

- In-app (HSBC HK App): Gamified “Risk Map” showing live allocation
- WeChat Mini Program: Educational content (“Your risk changes 47x per week – does your insurance?”)
- HSBC Life agents: Commission for referring customers to the digital upgrade (no underwriting work)

5. Expected Outcomes & Impact

Business Impact:

Metric	Launch Year 1	Scaled Year 3
Customer adoption	5–8% of HSBC retail customers	15–20% of HSBC retail customers
Usage intensity	50–100 micro-events per user/year	120–150 micro-events per user/year
Revenue per user	\$10–20	\$24–30
Incremental revenue	\$15–25M	\$120–150M+
Gross margin	>85%	>90%
Customer retention	85–90%	90%+
Cross-sell to long-term policies	15–20%	25–30%

Key drivers:

1. Near-zero marginal servicing cost, enabled by agentic AI and no traditional underwriting.
2. Built-in distribution through the HSBC app, which limits customer acquisition spend.
3. Incremental to existing life products, supporting higher wallet share rather than product cannibalization

Bottom line: A high-margin, recurring revenue stream that monetises real-time risk without cannibalising HSBC's core insurance book.

Customer Impact:

1. Improved protection, as estimated 40% reduction in underinsurance events (e.g., uninsured travel medical).
2. 90% of pilot users would prefer fluid coverage to fixed policies.
3. Financial resilience as micro-boosts (\$3-\$5) replace expensive last-minute travel insurance.

Social / ESG Impact: Directly addresses the UN's SDG 1.4 (access to economic resources) and SDG 3.8 (financial protection against health risks). By closing the protection gap for gig and mobile workers, Micro Protection Fluid reduces reliance on social safety nets and family borrowings, a core ethical obligation cited by the majority of insurance executives.

6. Potential Risk and Mitigation Strategy

Risk	Impact	Mitigation Strategy
Regulatory Approval Delays	High: Could delay GBA launch by 6+ months	Pre-engage HKIA/China regulators via HSBC's existing Wealth Management Connect pilots; start with HK-only parametric insurance (no claims adjusters needed)
AI Decision Errors	Medium: Wrong allocation could lead to undercoverage claims	Floor-and-ceiling model ensures minimum guaranteed coverage; human override via app; start with manual guardrails in Phase 1 pilot
Data Privacy Concerns	High: Customers wary of transaction data usage	Full transparency dashboard shows exactly what data triggers what coverage; opt-in only with granular controls; CCPA compliant by design
Low Customer Adoption	Medium - If users don't trust "black box" AI	Gamified Risk Map + educational WeChat content; 90-day free trial for premier customers; agent referral commissions
Tech Integration Failure	Low: Card auth delays or Merchant Category Code misclassification	Fallback to baseline coverage if AI fails; Phase 1 uses only existing HSBC transaction data (no 3rd parties); sub-second response tested in sandbox

7. Conclusion

Micro Protection Fluid transforms static insurance into a dynamic, agentic shield that perfectly matches real-time risks, delivering unmatched customer value through HSBC's existing app and data infrastructure while generating high-margin GBA revenue with minimal execution risk.

8. References

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